

SKEMPI 2.0 $\Delta\Delta G$ Predictor — Limitations & Future Work

Model & Structure

- Predict the mutant 3D structure explicitly (AlphaFold3 / ESMFold / IgFold) instead of relying only on an embedding-space shift, to capture real conformational change.
- Compare wild-type vs. predicted mutant geometrically in 3D (e.g. interface RMSD / displacement) rather than only in embedding space.
- Test other pretrained structure/sequence encoders in place of SaProt (e.g. ESM-2, AlphaFold-based embeddings, AntiFold) to see whether a different backbone improves on the current results.

Data Currently Excluded

- Integrate the 59 “ineq” (bounded) and 79 “n.b.” (no binding) entries via a hinge-style loss term instead of discarding them.
- Use same-mutation multi-measurement spread (currently deduplicated away) as a per-entry noise/uncertainty signal.
- Outlier removal — not yet implemented, approach undecided.

Metadata & Auxiliary Signals

- Richer metadata inputs beyond temperature: solvent, pH, buffer, experimental method.
- k_{on}/k_{off} (1,844 entries) and $\Delta H/\Delta S$ (443 entries) as auxiliary multi-task targets — noisier (58–83% cross-measurement agreement vs. 84% for $\Delta\Delta G$), would need careful weighting.

Loss & Metrics

- Hard-example / tail-weighted loss — plain MAE underweights rare, large-effect mutations (the 5-bin results already show higher error on the destabilizing/stabilizing tails).
- Loss-term balance check once the hinge term is reintroduced (unconfirmed whether MAE and hinge terms are properly balanced).
- Metrics broken out jointly by all three stratification axes at once.

Stratification

- Balance mutation location (antibody- vs. antigen-side) across the train/val split.
- Alanine-scanning (X→A, >50% of single mutations) stratification, to avoid an “alanine-ness” shortcut.
- Confirm the homology-based dedup uses SKEMPI’s own homology annotation field rather than a self-computed equivalent.